TECHNOLOGY SOLUTIONS UK LTD
part of HID

3138 *BLUETOOTH*® UHF RFID RAIN READER

HIGH PERFORMANCE, LONG RANGE UHF RFID READING WITH EPOP-LOQ® CONNECTIVITY

Supports GEN2X

RAIN RFID

Bluetooth®

Made for iPhone | iPad





Latest Hardware Advances



TSL®'s popular 2128/2128P series has been updated with new, performance-boosting internal hardware and the market's most advanced security features. The inclusion of a 'Power Handle' increases battery capacity, which combined with more efficient hardware provides longer running times than ever before. A Hot-Swap accessory provides the ability for continuous, 24/7 operation of the handheld reader.

Using the latest generation of Impinj silicon, read rates have been boosted to up to 1200 tags per second. The 3138 Reader has been given a faster ePop-Loq® mode than previous TSL readers to accommodate the higher tag data throughput. On reader de-duplication for 1 million transponders prevents overwhelming the connected host with information.

Multiple tag operations per tag are possible during an inventory scan - read multiple areas of each tag, or write to or even lock the tag.

Support has been added for HID and other sensor transponders (temperature, strain, humidity, acceleration) and a demo app is provided.

Designed to read and write to EPC Class 1 Gen 2 and EPC Class 1 Gen 2V2 tags, the 3166 can also be configured with class leading high performance 2D data scanning to bring unparalleled data collection capabilities to any host it is connected to.

Flexible charging options



The 3138 Reader's Power Handle contains integrated charging electronics, meaning that the Power Handle can be charged as a stand-alone unit using any 30W+ USBC PD Power Supply. Buy a spare Power Handle and always have a fully charged battery on standby, for non-stop scanning! Host devices attached to the Reader using an ePop-Loq mount (see page 6) can be charged at the same time, providing a single point charging solution.

Quick Configuration with Command Scan



3138 Readers with the 2D Imager option can use TSL's patent-pending 'Command Scan' feature to scan special barcodes to configure the Reader. This allows lightning fast Reader configuration, along with on-the-fly adjustment of Reader parameters, opening up a world of improved workflow scenarios. Some barcode examples can be viewed in this document.

Batch Mode

Transponder EPC readings can optionally be stored on the embedded 32GB Micro SD card, meaning that the 3138 Reader can be used independently of a host device. Up to 500 million* EPCs can be stored - all date and time stamped by the on-board Real Time Clock.

Market's Most Advanced Security



The 3128 Reader is future regulation/Cyber Security Acts ready. It features Tag Authentication, Secure Data Storage, Encrypted Communication, Data Acquisition Authenticity checks, Anti-Counterfeiting Measures and Personnel Authentication.

Features:

Up to 6x faster tag read rate
Gathers RFID data at a blistering pace - up to 1200 tags per second.

Multiple RF Profiles
- high sensitivity
- maximum read rate
- dynamic switching between modes

ETSI Upper Band Option

Bulk encoding feature

***Bluetooth*® improvements**
Around a 2x speed boost for Bluetooth Classic in iOS. Bluetooth Smart, also known as Bluetooth Low Energy, provides a modern, secure link to even more devices, no need for MFI app approval, and out-of-band battery status and device information).

Hardware and OS Independence
Use standalone, or with a wide variety of Bluetooth wireless technology enabled host devices including smartphones, tablets, Enterprise Handhelds and PC's. Operates with Android, iOS, Linux, Mac, Windows.

Backward Compatible
TSL focus on providing long term support of their RFID Readers and minimal design disruption. New TSL Readers are compatible with existing software. Software only needs to be updated if wanting to make use of newly added features.

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3138 SPECIFICATIONS*

Physical and Environmental Characteristics

Dimensions:	158 x 98 x 170 mm (LxWxH).
Weight:	550g (including Power Handle)
User input:	Trigger button.
User feedback:	Speaker, vibration motor, LED - user configurable.
Power:	Rechargeable 25.4Wh, 7.3V Lithium Polymer battery pack
Minimum operating time[1]:	Light use[2]: 20 hrs Moderate use[3]: 11 hrs Heavy use[4]: 6.5 hrs (preliminary estimates)
Supported USB Power Delivery (PD) Profiles:	Fixed PDOs at 5V/3A, 9V/3A, 15V/3A (up to 45W max)
Enclosure materials:	Polycarbonate.

Performance Characteristics

RFID engine:	TSL custom module with embedded Impinj E710.
Communication protocols:	TSL ASCII 3 parameterised command set.
Security features:	HID OMNIKEY® Security Chip
Memory:	Embedded 32GB[5] storage memory - store up to 500 million date and time stamped EPCs
Compatible host devices (*Bluetooth*®):	Any *Bluetooth*® Host[6] supporting the Serial Port Profile (SPP), Human Interface Device (HID), BLE with HOG (HID over GATT) or BLE with SOG (Serial over GATT). See Bluetooth Mode Comparison.
Compatible host devices (USB):	Any USB host with USB CDC support (Windows, Linux, Mac, Android).

Environmental

Operating temp.:	-10°C to 50°C (14°F to 122°F).
Charging temp.:	5°C to 40°C (41°F to 104°F).
Storage temp.:	Less than 1 month at -20°C to +45°C (-4°F to 113°F). Less than 6 months at -20°C to +35°C (-4°F to 95°F).
Humidity:	5% to 85% non-condensing.
Drop spec:	Multiple drops to concrete: 4 ft./1.2 m ambient, 3ft / 0.9m across the operating temperature range.
Tumble:	1000 0.5 metre tumbles at room temperature (2000 cycles).
Environmental Sealing:	IP54.
Electrostatic Discharge (ESD):	± 15kVdc air discharge; ± 8kVdc contact discharge.
MIL-STD 810F:	Meets and exceeds applicable MIL-STD 810F for drop, tumble and sealing.

RFID Performance

Standards supported:	EPC Class 1 Gen 2 v2
Nominal read range[7]:	Up to 9 m (29.5 ft).
Nominal write range[7]:	Up to 4 m (13.1 ft).
Field:	150-degree forward facing (approx.)
Antenna:	Right Hand Circularly Polarised with optional 2D scanner.
Frequency Range:	EU: 865-868MHz, 916-919MHz US: 902-928MHz
Maximum Output Power:	Up to 30 dBm (region dependent) + 4.0 dBiC Antenna.

Barcode Scanning

Optional 2D barcode engine:	Optional TSL custom 2D Barcode Scan Engine module.
Sensor resolution:	1280 x 960 pixels, rolling shutter
Field of view:	Horizontal: 44.5°, vertical: 33.5°
Focal distance:	From front of engine: 15.24 cm (6 in.)
Aiming LED:	Green LED
Illumination:	1 warm white LED
Symbologies supported:	1D: All major codes 2D: PDF417, MicroPDF417, Composite, RSS, TLC-39, Datamatrix, QR code, Micro QR code, Aztec, MaxiCode Postal Codes: US PostNet, US Planet, UK Postal, Australian Postal, Japan Postal, Dutch Postal (KIX).

Ranges[8]:

Barcode	Near	Far
5 mil Code 39	6.1 cm	24.1 cm
5 mil Code 128	7.1 cm	22.9 cm
6.67 mil PDF 417	6.1 cm	20.3 cm
10 mil DataMatrix	7.4 cm	21.6 cm
100% UPCA	4.6 cm	49.5 cm
15 mil QR	3.0 cm	29.2 cm
20 mil QR	3.0 cm	35.6 cm

Communication

Bluetooth®:	*Bluetooth*® v4.2 compliant (v5.1 compatible)
Bluetooth GATT Services:	• Device Information Service • Battery Service • HID over GATT • Serial over GATT (TSL)
Bluetooth frequency range:	2.4 - 2.4835 GHz.
Bluetooth profiles:	SPP Profile, HID Profile, Apple iAP2, Bluetooth Low Energy.
Bluetooth range[9]:	Up to 100m.
Bluetooth pairing:	Simple Secure Pairing, NFC OOB Pairing.
Physical host device connection	Direct connection via ePop-Loq® cases (separate purchase).

[1] Minimum operating time figures are based on new units that have been stored, charged and operated within the stated Environmental Specifications. Units stored over 3 months must be recharged every 3 months. Number of transponders in the environment affects minimum operating time.

[2] Light Use: Continuous RFID inventories for 20s of every 120s

[3] Moderate Use: Continuous RFID inventories for 10s of every 30s

[4] Heavy Use: Continuous RFID inventories for 59s of every 60s

[5] Up to 256GB microSD card storage supported. 32GB fitted as standard

[6] Compatible *Bluetooth*® stack required in the Host device

[7] Tag Read/Write performance is dependent on tag type, items tagged, number of tags in the field and other radio and environmental factors

[8] Artificial lighting can affect scanning performance

[9] Open field

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3138 SPECIFICATIONS*

Peripherals and Accessories

External interface:	USB-C connector or custom 8-Way Power-Pin connector (for use with docking cradle)
Power supply management	Separate purchase (order USBC-PSU-KIT45)
Other accessories available:	See page 5 for available accessories See page 6 for device mounts for a variety of handheld terminals.

Regulatory

Regions	EU (CE), USA (FCC), Canada (see page 4 for details)
FCC ID	Pending
IC	Pending
EMC	EN 55032: 2015+A11:2020, Class A EN 6100-3-2: 2014, Class A EN IEC 61000-3-2: 2019+A1:2021, Class A EN 61000-3-3: 2013+A1:2019+A2:2021 EN 55035: 2017+A11:2020 EN 301 489-1 V2.2.3 (2019-11) EN 301 489-3 V2.3.2 (2023-01) EN 301 489-17 V3.2.4 (2020-09) 47 CFR FCC Part 15, Subpart B, Class A ANSI C63.4-2014 ANSI C63.4a-2017 ICES-003: 2020 Issue 7, Class A ICES-Gen: 2018 Issue 1 +A1:2021 ANSI C63.4-2014 amended as per ANSI C63.4a-2017
RF	EN 302 208 V3.3.1 (2020-08) 47 CFR FCC Part 15, Subpart C (Section 15.247) ANSI C63.10-2013 Canada RSS-247 Issue 3, August 2023 Canada RSS-Gen Issue 5, Amendment 2, February 2021 ANSI C63.10-2013
RF Exposure	EN 50566:2017 EN 62209-2:2010/A1:2019 IEC 62209-2:2010/AMD1:2019 EN 50663:2017 EN 62479:2010 EN 50364:2018 EN 62369-1:2009 IEEE Std 1528:2013 KDB 865664 D01 v01r04 KDB 865664 D02 v01r02 KDB 447498 D01 v06 ISED RSS-102 Issue 5:2015/AMD1:2021 IEC/IEEE 62209-1528:2020
Electrical Safety	IEC 62368-1:2018 UL 62368-1:2019 CSA C22.2 No. 62368-1:19
Environmental	**2011/65/EU (RoHS 2)** Restriction of the use of certain Hazardous Substances in electrical and electronic equipment **2015/863 (RoHS 3)** Amendment to Annex II of 2011/65/EU



USB-C connector option



8-Way Power Pins option, for use with Docking Cradle

Warranty

The TSL 3138 Reader is warranted against manufacturing defects for a period of one year (12 months), provided the product remains unmodified and is operated under normal and proper conditions.

Full warranty information can be downloaded from the TSL website at www.tsl.com/warranty

Maintenance Programs

To enhance and extend your warranty, consider purchasing a CompleteCare Maintenance Program for one-stop coverage with extensive customer support.

- 1, 3 & 5 year terms available
- Advance exchange with comprehensive cover option (including accidental damage cover)

CompleteCare

Terms

The *Bluetooth*® word mark and logos are registered trademarks owned by *Bluetooth* SIG, Inc. and any use of such marks by Technology Solutions UK Ltd is under license. Other trademarks and trade names are those of their respective owners.

"Made for iPod," "Made for iPhone," and "Made for iPad" mean that an electronic accessory has been designed to connect specifically to iPod, iPhone, or iPad, respectively, and has been certified by the developer to meet Apple performance standards. Apple is not responsible for the operation of this device or its compliance with safety and regulatory standards. Please note that the use of this accessory with iPod, iPhone, or iPad may affect wireless performance.

iPad, iPhone, iPod and iPod touch are trademarks of Apple Inc., registered in the U.S. and other countries.

TSL RFID Apps

These TSL RFID Demo apps have all been written using our free Software Development Kits:

- RFID Sensor Tag Explorer https://www.tsl.com/apps/sensor-tag-explorer
- RFID Scan Scan Write www.tsl.com/apps/rfid-scan-scan-write
- RFID Tag Finder www.tsl.com/apps/rfid-tag-finder
- RFID Web Wedge www.tsl.com/apps/rfid-web-wedge
- RFID Explorer www.tsl.com/apps/rfid-explorer
- TSL Reader Configuration www.tsl.com/apps/tsl-reader-configuration
- RFID Inventory https://www.tsl.com/apps/rfid-inventory

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3138 PART NUMBERS

Countries*	Part Numbers	Operating Frequency
ETSI Lower Band Albania, Andorra, Austria, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Falkland Islands, Finland, France, French Guiana, Georgia (Licence Required), Germany, Greece, Greenland, Guernsey, Guadeloupe, Hungary, Iceland, Ireland, Italy, Jersey, Latvia, Liechtenstein, Lithuania, Luxembourg, Macedonia, Malta, Martinique, Monaco, Montenegro, Netherlands, Norway, Poland, Portugal, Qatar, Romania, Saudi Arabia, Slovakia, Slovenia, Spain, Sweden, Switzerland, United Kingdom	USB-C connector, 2D barcode imager: 3138-ES1-EU USB-C connector, no barcode imager: 3138-EX1-EU 8-Way Power Pins, 2D barcode imager: 3138-ES2-EU 8-Way Power Pins, no barcode imager: 3138-EX2-EU	• ETSI Lower Band • Operating frequency: 865 – 868 MHz
ETSI Upper Band Albania, Austria, Belgium*, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Greece, Hungary, Iceland, Ireland, Italy, Liechtenstein*, Lithuania, Luxembourg, Macedonia, Malta, Moldova, Montenegro, Norway, Poland, Portugal, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland*, Türkiye, United Kingdom *Limited implementation	USB-C connector, 2D barcode imager: 3138-AS1-EU USB-C connector, no barcode imager: 3138-AX1-EU 8-Way Power Pins, 2D barcode imager: 3138-AS2-EU 8-Way Power Pins, no barcode imager: 3138-AX2-EU	• ETSI Upper Band • Operating frequency: 916 - 919 MHz • Each country listed **may have further requirements or restrictions** - for more information, visit www.tsl.com/uhf-frequency-allocations/

		USB-C connector:	8-Way Power Pins:	
FCC USA, Canada, Ecuador, Guam, Guatemala, Northern Mariana Islands, Puerto Rico	2D barcode imager:	3138-AS1-US	3138-AS2-US	• FCC • Operating frequency: 902 – 928 MHz
	No barcode imager:	3138-AX1-US	3138-AX2-US	
Australia	2D barcode imager:	3138-AS1-AU	3138-AS2-AU	• Operating frequency: 920 - 925 MHz
	No barcode imager:	3138-AX1-AU	3138-AX2-AU	
Brazil	2D barcode imager:	3138-AS1-BR	3138-AS2-BR	• Operating frequency: 902 – 907, 915 – 927 MHz
	No barcode imager:	3138-AX1-BR	3138-AX2-BR	
Colombia	2D barcode imager:	3138-AS1-CO	3138-AS2-CO	• Operating frequency: 915 – 927 MHz
	No barcode imager:	3138-AX1-CO	3138-AX2-CO	
New Zealand	2D barcode imager:	3138-AS1-NZ	3138-AS2-NZ	• Operating frequency: 920 - 925 MHz
	No barcode imager:	3138-AX1-NZ	3138-AX2-NZ	
Vietnam	2D barcode imager:	3138-AS1-VN	3138-AS2-VN	• Operating frequency: 919 – 922 MHz • Power limited to 31 dBm (EIRP)
	No barcode imager:	3138-AX1-VN	3138-AX2-VN	

*If you are interested in purchasing for a country/region that is not listed above, please contact enquiries@tsl.com for assistance.

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3138 PART NUMBERS

Accessories



USBC-PSU-KIT45 - 45W USB-C PSU with Power Delivery (PD)

For charging 3138 Readers fitted with a USB-C Power Handle. Also for use with Docking Cradle Kit. Includes UK, US, EU and AU plug adapters.



3138-CRD-01-KIT - Docking Cradle Kit for 3128/3138 RFID Readers, includes USB-C cable and spacer clip

For charging 3128/3138 Readers fitted with an '8-Way Power Pins' Power Handle. Requires a 30W or higher USB-C PSU with Power Delivery (not included, order USBC-PSU-KIT45).



31PH-01 - 31x8 Power Handle (USB-C connector) – includes USB-C cable

Enjoy continuous operation of the 3138 Reader, by keeping a spare Power Handle on charge. Includes USB-C cable.



31PH-02 - 31x8 Power Handle (8-way Power Pins)

Enjoy continuous operation of the 3138 Reader, by keeping a spare Power Handle on charge. For use with Docking Cradle Kit.



31-HS-01 - 31x8 Hot-Swap Accessory

This optional upgrade allows you to keep the reader ON whilst changing the Power Handle, for non-stop RFID scanning. Simply swap Power Handles, and pull the trigger to continue your inventory - without interrupting your process or losing connection to the host device.

For use with both USB-C and 8-way Power Pin variants.

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MOUNTING A HOST DEVICE ON THE UHF READER

You can physically connect Enterprise Hand-Held Terminals to your UHF RFID Reader using ePop-Loq® mounts. These custom made mounts can provide direct USB charge and data connections (when available) between the Reader and host device. To enquire about ePop-Loq mounts for devices not listed below, please contact enquiries@tsl.com.


Datalogic Memor 10/11




Datalogic Memor 12/17




Datalogic Memor 30/35




Honeywell CT40/45/45XP/47




Honeywell CT50/60




Honeywell D75e




Honeywell EDA50/51




Janam XT30/40




Zebra TC21/26




Zebra TC22/27




Zebra TC51/52/56/57




Zebra TC70/72/75/77



Device Mount	Part Numbers
ePop-Loq Mount for Datalogic Memor 10/11	2108-A-EPL-CASE
ePop-Loq Mount for Datalogic Memor 12/17	2118-A-EPL-CASE
ePop-Loq Mount for Datalogic Memor 30/35	2117-A-EPL-CASE
ePop-Loq Mount for Honeywell CT40, CT45/ CT45XP and CT47	2103-A-EPL-CASE-V3
ePop-Loq Mount for Honeywell CT50/CT60	1192-A-EPL-CASE
ePop-Loq Mount for Honeywell D75e	1190-A-EPL-CASE
ePop-Loq Mount for Honeywell EDA50	1195-C-EPL-CASE
ePop-Loq Mount for Honeywell EDA51	1217-C-EPL-CASE
ePop-Loq Mount for Janam XT30/XT40	ATC-XT-01 (order through Janam Technologies)
ePop-Loq Mount for Xiaomi Mi A2	2106-A-EPL-CASE
ePop-Loq Mount for Zebra TC20/TC25	2101-A-EPL-CASE
ePop-Loq Mount for Zebra TC21/TC26	2120-A-EPL-CASE
ePop-Loq Mount for Zebra TC22/TC27	2119-A-EPL-CASE
ePop-Loq Mount for Zebra TC51/52/56/57	1199-A-EPL-CASE
ePop-Loq Mount for Zebra TC70/72/75/77	1198-A-EPL-CASE



Charge the 3138 Reader *and* host device, using ePop-Loq mounts)



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ABOUT TSL

Technology Solutions UK Ltd (TSL®), part of HID, is a leading manufacturer of high performance mobile RFID readers used to identify and track products, assets, data or personnel.

For over two decades, TSL has delivered innovative data capture solutions to Fortune 500 companies around the world using a global network of distributors and system integrators. Specialist in-house teams design all aspects of the finished products and software ecosystems, including electronics, firmware, application development tools, RF design and injection mould tooling.

TSL is an ISO 9001:2015 certified company.

ISO 9001: 2015

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ABOUT HID

HID powers the trusted identities of the world's people, places and things. We make it possible for people to transact safely, work productively and travel freely. Our trusted identity solutions give **people** convenient access to physical and digital **places** and connect **things** that can be identified, verified and tracked digitally. Millions of people around the world use HID products and services to navigate their everyday lives, and billions of things are connected through HID technology. We work with governments, educational institutions, hospitals, financial institutions, industrial businesses and some of the most innovative companies on the planet. Headquartered in Austin, Texas, HID has over 4,000 employees worldwide and operates international offices that support more than 100 countries. HID is an ASSA ABLOY Group brand.

For more information, visit www.hidglobal.com.

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